R56: complete divisor, infinity points and rational uniformizers

The actual source now has a complete A=1+c principal divisor. Both infinity fibers split into one ramified degree-7 and one unramified branch, each with residue field F4. Explicit rational uniformizers bind the normalization to source coordinates. The function lambda/A has certified simple poles at all four finite A zeros, giving a concrete prescribed-pole RR construction.

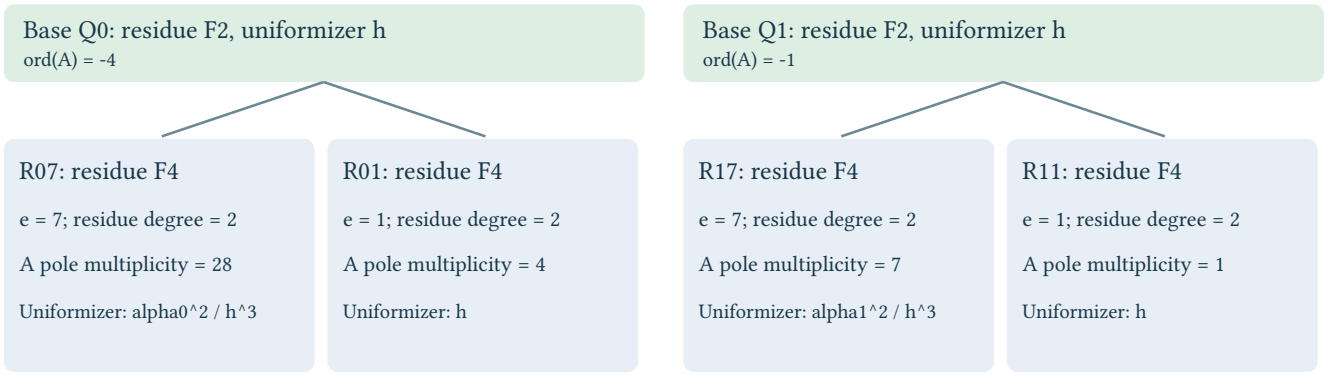
Exact base chart and pole orders

For the actual source T=lambda, put h=1/z, $W=h^2V$ and $s=h^4$lambda. Full rational chart identities transform the base to $W^2+W=h+h^3$ and bind $R=(1+z)V+z^3$, b=zR(R+1), c=z(R+1), A=1+c. The two simple residue roots W=0,1 lift uniquely. Exact telescoping and 128-jets give:

Base point	R	b	c	A	Delta
Q0	−3	−7	−4	−4	−37
Q1	2	1	−1	−1	−1

All entries are exact orders. Source: infinity56-02. In particular b has a boundary zero at Q1, which must be included when evaluating b denominators.

Complete source fibers at infinity



Each fiber has degree 2×7+2×1=16. These are closed points over F2, with residue degree 2. Both F4 conjugate residue charts are checked explicitly.

The first Newton faces were repeated eighth powers. One further source-bound translation resolves each cluster: positive-root segments have lengths 1 and 7, with root orders 2 and 4/7 over Q0, 1 and 4/7 over Q1. The degree-7 segment has full reduced denominator 7; the separable complete-local-field Newton theorem proves irreducibility and total ramification. The other segment is linear. Original-polynomial reconstruction and tracked precision pass at all stages: precision 120 over Q0, 72 over Q1. Sources: refine56-01/check56-01.

Actual uniformizers and the complete principal divisor

Source formulas for normalized local parameters

Define the following F2 rational coordinate functions:

```
alpha0=h^8*lambda^2+(h^4+h^5)*lambda+1+h+h^2
alpha1=h^2*lambda^2+(h+h^2)*lambda+1+h^2
```

In each ramified chart the exact source identity is

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alpha_i=h*t*(1+h+h*t); ord(h)=7; ord(t)=4
```

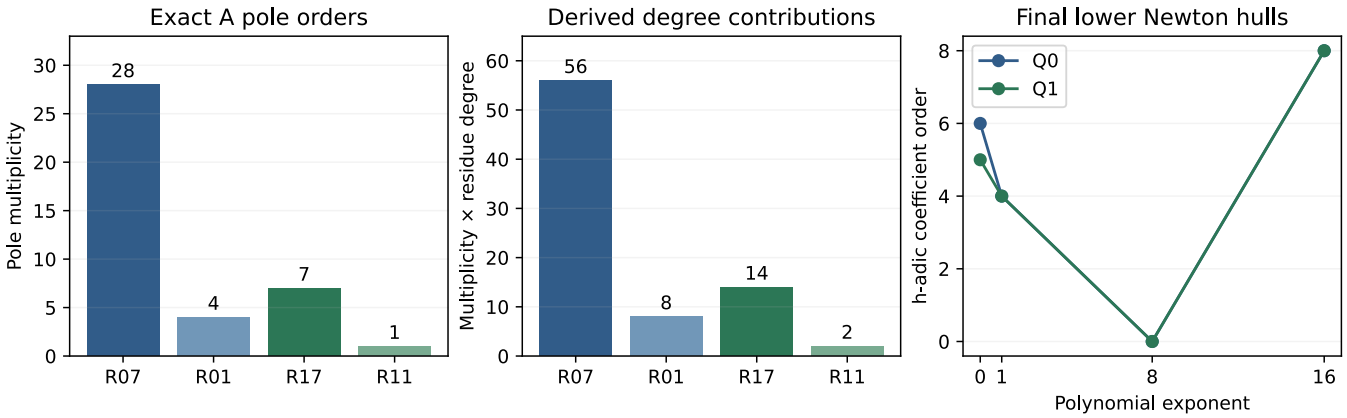
The last factor has residue 1. Hence alpha_i has order 11 and sigma_i=alpha_i²/h³ has order 22-21=1. These are rational uniformizers at R07 and R17. At R01 and R11, h is a uniformizer. The normalized completed local rings are F4 power-series rings in those parameters. This does not identify a smaller coordinate-image subring with a monomial ring.

All zeros and poles

R55 gives four reduced degree-20 zeros P1,...,P4. Their local maximal ideals are generated by A, so the Noetherian local domains are DVRs: normalization adds no branch or multiplicity there. The complete divisor is

```
div(A)=P1+P2+P3+P4 -28*R07-4*R01-7*R17-R11
```

Both zero and pole degrees are 80. Lambda has exactly the same pole divisor. Unit56-01 checks the actual lambda and V values on the entire degree-80 zero quotient, the border equation, and full Bezout products for lambda and b. Both are units there; a changed inverse is detected. Therefore lambda/A has exactly simple poles at the four Pi and no poles at infinity.



Multiplicities and Newton vertices are exact native records; degree bars are labelled derived products with residue degree 2. No sampling uncertainty.

Multiplication by A identifies L(P1+...+P4) with L(the pole divisor of A). Next match the retained complete 68-section ambient RR divisor to these actual points, then impose all local constraints to construct the full prescribed space.

Proof fidelity, failures and reproducible evidence

Four audited Lean implications

All four authored Infinity56 declarations compile and pass separate axiom audit, zero failing and zero trusted. They cover jet-order stability, exact chart scaling, pole pullback, and the full-denominator ramification/residue conclusion. Actual arrays, Hensel error bounds, local-field Newton facts and normalization/DVR interpretation remain explicit external premises. See PROOF_MAP.md. No sorry, native_decide or new axiom is used.

Full terminal workflow ledger

Workflow	Status	Receipt
check56-01	accepted	exit 0; 0 error lines
infinity-lean56-01	accepted	compile 0; audit 0; four declarations
infinity56-01	rejected	exit 0; 14 error lines
infinity56-02	accepted	exit 0; 0 error lines
refine56-01	accepted	exit 0; 0 error lines
unit56-01	accepted	exit 0; 0 error lines

Five workflows accepted, one rejected. The first infinity attempt failed an indexed-list mapping after its base prefix; its late marker is excluded. The corrected whole-list mapping passes. Integer-division and redefinition warnings are non-error diagnostics. All 35 executed historical byte pins match Git. The fresh local checker independently reconstructs every final chart and checks all Newton secant inequalities, coordinate identities and degree accounting, rather than relying only on the original hull algorithm.

Limits, preservation and sources

Native: fixed seed 56, 4 GiB virtual, 1,200 seconds, one Singular/FLINT thread. Lean 4.34.1 / mathlib d13f23b, -j1/-M4096, separate 1,200-second compile and audit caps, no virtual cap. Producer hostname and CPU times were not separately recorded; native UTC wall stamps are retained. No comparative speed claim.

All six terminal run directories survived the host transition. Only the missing metadata index was regenerated. No accepted producer was repeated. The original executed checkpoint helper is retained separately; a published main merge changed relay packaging outside its gate function. No managed Diem/freezer owner, research EKS job or new Conductor claim was started.

References checked: retained R50/R51 RR algebra, R52 source field, R53 base coordinates, R55 quotient; Stacks04GM Hensel, Stacks00PD DVR criterion, Stacks0C1B valuation/ramification; Garrett, Newton polygons, pages 1-2. The source-specific geometry is derived from the certificates above.

Next construction and preserved lanes

The retained source already has genus 61 and a complete 68-section RR space of degree 128. Match its exact R27 cusp divisor to the new points; after certifying inclusion, impose all degree-48 local constraints. Do not assume that ambient divisor equals four times the z-pole divisor. Expand all source basis functions and nu in the actual uniformizers, then test modular maps with complete denominator, descent and intended order-11 action checks. Preserve proper polarization, separate incidence saturation, node187 coupled divisors, arbitrary reverse classes, and the larger degree-112/991 lane.